PROJECT DATA SHEET

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JOB/PROJECT : MULTI FAITH CENTRE, TEMPORARY WORKS

JOB/PROJECT NO : C8676 **SHEET NO :** COVER

GENERAL :

CLIENT : WILLMOTT DIXON CONSTRUCTION LTD

ARCHITECT : N/A

AUTHORITY : EXETER CITY COUNCIL

JOB/PROJECT DESCRIPTION : HERAS FENCING

PROPOSED USE: SITE PERIMETER FENCING

OVERALL SIZE : AS CALCS **No. OF STOREYS :** N/A

TYPE OF SOIL INVESTIGATION : NONE

BEARING STRATA : N/A

TYPE OF FOUNDATION : N/A

MAXIMUM SAFE BEARING PRESSURE N/A kN/m²

STRUCTURAL MATERIALS USED : PROPRIATORY COMPONENTS

STANDARDS AND CODES USED IN DESIGN :

General actions: Densities, self weight & imposed loads	BS EN 1991-1-1	☒
General actions: Actions on structures exposed to fire	BS EN 1991-1-2	☐
Snow Loads	BS EN 1991-1-3	☐
Wind Loads	BS EN 1991-1-4	☒
Concrete	BS EN 1992 (EC2)	☐
Steelwork	BS EN 1993 (EC3)	☐
Timber	BS EN 1995 (EC5)	☐
Masonry	BS EN 1996 (EC6)	☐
Others	TW:2012-1 Hoardings A Guide to Good Practice April 2021	☒
		☐
		☐

AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: H01— H05 inclusive	A.C. HEARD	29/08/23

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Calculations

Contract	MULTI FAITH CENTRE TEMPORARY WORKS	Job No.	C8676	Sheet No.	H01.
		Designed by	D.P.H	Date:	AUG-23

Scheme Proposal

It is proposed to construct a Heras fence around an area at Exeter University.

The fencing will be positioned at various orientations. As such, design of members will be based on the worst case wind loading for this location, i.e. 300° from North.

General wind loading has been considered as there is no drop or pedestrian walkway adjacent to the Heras panels. The 0.74 kN/m guarding and nominal wind ($0.2kN/m^2$) are therefore not applicable in this instance. The solidarity ratio of the fence and debris netting has been taken as 30%

The design life of the fencing is less than two years.

/MetSPEC EURO/®
DESIGN SUITE

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Job No.: C8676
Designer: DePoH
Date: 25/08/2023
Registered Details:-
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Wind Assessment to BS EN 1991-1- 4

Data Entry:-

Site Altitude	**68.000 m**	**Reference Height (Z)**			**Size Effect Dimension (b + h)**		
$V_{b,map}$	**23.000 m/s**	**Roof**	**2.000**	**m**	**Roof**	**0.000**	**m**
Seasonal Factor (C,season)	**1.000**	**Side Walls**	**2.000**	**m**	**Side Walls**	**0.000**	**m**
Probability Factor (C,prob)	**1.000**	**Gables**	**2.000**	**m**	**Gables**	**0.000**	**m**
Site ID							

Dynamic Pressure Results

Wind Direction (deg)		**0**	**30**	**60**	**90**	**120**	**150**	**180**	**210**	**240**	**270**	**300**	**330**
Direction Factor C,dir		**0.78**	**0.73**	**0.73**	**0.74**	**0.73**	**0.80**	**0.85**	**0.93**	**1.00**	**0.99**	**0.91**	**0.82**
Orography Factor Co		1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Effective Height (h-hdis) m	**Roof**	2.000	0.800	0.800	0.800	0.800	0.800	0.800	0.800	0.800	0.800	2.000	2.000
	Sides	2.000	0.800	0.800	0.800	0.800	0.800	0.800	0.800	0.800	0.800	2.000	2.000
	Gable	2.000	0.800	0.800	0.800	0.800	0.800	0.800	0.800	0.800	0.800	2.000	2.000
Altitude Factor C,alt	**Roof**	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068
	Sides	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068
	Gable	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068	1.068
Roughness Factor Cr	**Roof**	0.708	0.420	0.412	0.404	0.417	0.421	0.414	0.417	0.421	0.423	0.716	0.709
	Sides	0.708	0.420	0.412	0.404	0.417	0.421	0.414	0.417	0.421	0.423	0.716	0.709
	Gable	0.708	0.420	0.412	0.404	0.417	0.421	0.414	0.417	0.421	0.423	0.716	0.709
Exposure Factor Ce	**Roof**	1.387	0.940	0.904	0.870	0.930	0.946	0.918	0.931	0.946	0.956	1.424	1.392
	Sides	1.387	0.940	0.904	0.870	0.930	0.946	0.918	0.931	0.946	0.956	1.424	1.392
	Gable	1.387	0.940	0.904	0.870	0.930	0.946	0.918	0.931	0.946	0.956	1.424	1.392
Vb,0 (m/s)	**Roof**	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564
	Sides	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564
	Gable	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564	24.564
Vb (m/s)	**Roof**	19.160	17.932	17.932	18.177	17.932	19.651	20.879	22.845	24.564	24.318	22.353	20.142
	Sides	19.160	17.932	17.932	18.177	17.932	19.651	20.879	22.845	24.564	24.318	22.353	20.142
	Gable	19.160	17.932	17.932	18.177	17.932	19.651	20.879	22.845	24.564	24.318	22.353	20.142
Vm (m/s)	**Roof**	13.559	7.527	7.386	7.352	7.481	8.268	8.648	9.530	10.339	10.283	15.996	14.274
	Sides	13.559	7.527	7.386	7.352	7.481	8.268	8.648	9.530	10.339	10.283	15.996	14.274
	Gable	13.559	7.527	7.386	7.352	7.481	8.268	8.648	9.530	10.339	10.283	15.996	14.274
Turbulence Intensity Iv	**Roof**	0.212	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.212	0.212
	Sides	0.212	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.212	0.212
	Gable	0.212	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.411	0.212	0.212
Peak Velocity Pressure qp (kN/m²)	**Roof**	0.301	0.173	0.167	0.165	0.171	0.209	0.229	0.278	0.327	0.323	0.419	0.334
	Sides	0.301	0.173	0.167	0.165	0.171	0.209	0.229	0.278	0.327	0.323	0.419	0.334
	Gable	0.301	0.173	0.167	0.165	0.171	0.209	0.229	0.278	0.327	0.323	0.419	0.334
[illegible]	**Roof**	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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Calculations

Contract	MULTI FAITH CENTRE TEMPORARY WORKS	Job No.	C8676	Sheet No.	H 03
		Designed by	D.P.H	Date:	AUG 23

DESIGN OF HERAS FENCING

Wind load:

Site location : SX 918 - 939

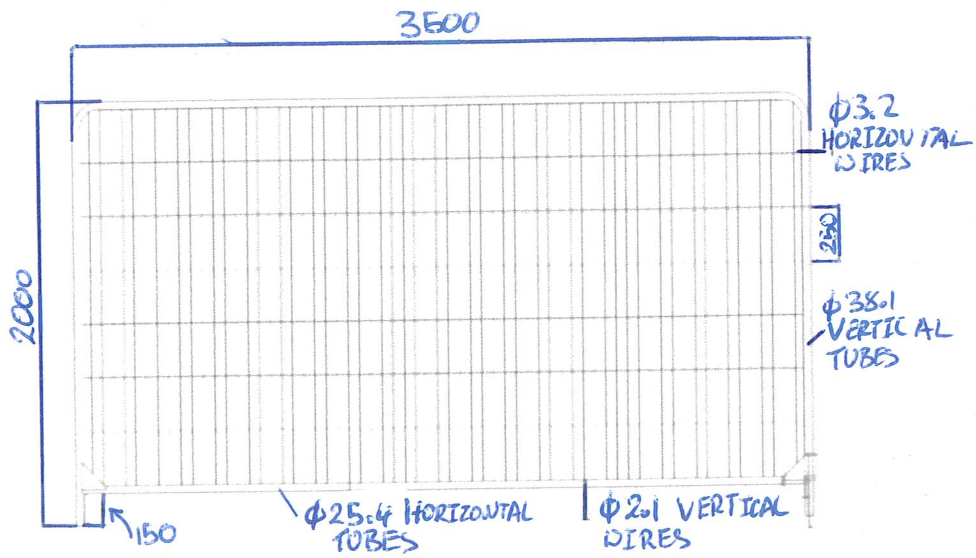
Site altitude : 68m

Peak velocity pressure is determined using MetSPEC (Page H02)

Peak velocity pressure, q_p : $0.419 kN/m^2$ (300 degrees from North)

Design life < 2 years ∴ Reduction coefficient $k = 0.77$ is applicable.

Fencing dimensions:



Netting

Table 4.5 – Aerodynamic force coefficients adopted for TG20 compliant scaffolds

From TG20:21

		Sheeting	High-permeability netting
Wind force normal to the cladding	$c_{f\perp}$	1.30	$\boxed{0.65}$
Wind force parallel to the cladding	$c_{f\parallel}$	0.10	0.17

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Calculations

Contract	MULI FAITH CENTRE TEMPORARY WORKS	Job No.	C8676	Sheet No.	H04
		Designed by	D.P.H	Date:	AUG 23

FORCES:

Assumed 30% Netting

$q_p \times k \times \ldots \times C_f$

$0.419 \times 0.77 \times 0.3 \times 0.65 = 0.063\,kN/m^2$ (CHAR) $\times 1.5$

$0.094\,kN/m^2$ (ULT)

moment $= 0.094 \times 3.5 \times 2.0 \times \frac{2}{2} = 0.658\,kNm$

Restoring moment about A: For layout refer to sheet 5.

MR =	Base	=	0.49	×	$\frac{0.75}{2}$	=	0.184 kNm
	Fence	=	0.15	×	0.68	=	0.102 kNm
	Balast base	=	0.44	×	0.51	=	0.224 kNm
			0.44	×	0.36	=	0.158 kNm
			0.44	×	0.21	=	0.092 kNm

$\Sigma = 0.76\,kNm > 0.66 \therefore$ Ok

Restoring moment about B:

MR =	Base	=	0.49	×	$\frac{0.75}{2}$	=	0.184 kNm
	Fence	=	0.15	×	0.07	=	0.011 kNm
	Balast base	=	0.44	×	0.239	=	0.105 kNm
			0.44	×	0.538	=	0.237 kNm
			0.44	×	0.389	=	0.171 kNm

$\Sigma = 0.71\,kNm > 0.66 \therefore$ Ok

$\therefore$ USE 49kg 750 × 210 × 120 FENCE FEET AND 3 No. 44kg HI-VIS HARAS BALAST BLOCKS, ARRANGED AS SHOWN ON PAGE H05

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		Designed by	DoPoH	Date:	AUG 23